

Sobel kernel — Sobel edge filter on the T1 fabric (3×3 worked example)

Steps run top → bottom. Each image pixel has its own colour, so you can track its position: Step 2 moves data sideways (within a row), Step 4 moves the SAME data up/down (between rows) using the identical vslidedown / vslideup — only the mode bit changed. Gx/Gy are sign-coloured (warm +, cool –).

